MWS5101, MWS5101A

256-Word x 4-Bit LSI Static RAM

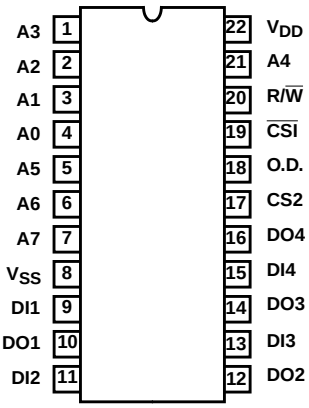
March 1997

Features

- **Industry Standard Pinout**
- **Very Low Operating Current . . . . . 8mA at V_{DD} = 5V and Cycle Time = 1μs**
- **Two Chip Select Inputs Simple Memory Expansion**
- **Memory Retention for Standby. . . . . 2V (Min) Battery Voltage**
- **Output Disable for Common I/O Systems**
- **Three-State Data Output for Bus Oriented Systems**
- **Separate Data Inputs and Outputs**
- **TTL Compatible (MWS5101A)**

Pinout

MWS5101, MWS5101A
(PDIP, SBDIP)
TOP VIEW



Description

The MWS5101 and MWS5101A are 256 word by 4-bit static random access memories designed for use in memory systems where high speed, very low operating current, and simplicity in use are desirable. They have separate data inputs and outputs and utilize a single power supply of 4V to 6.5V. The MWS5101 and MWS5101A differ in input voltage characteristics (MWS5101A is TTL compatible).

Two Chip Select inputs are provided to simplify system expansion. An Output Disable control provides Wire-OR capability and is also useful in common Input/Output systems by forcing the output into a high impedance state during a write operation independent of the Chip Select input condition. The output assumes a high impedance state when the Output Disable is at high level or when the chip is deselected by $\overline{CS1}$ and/or CS2.

The high noise immunity of the CMOS technology is preserved in this design. For TTL interfacing at 5V operation, excellent system noise margin is preserved by using an external pull-up resistor at each input.

For applications requiring wider temperature and operating voltage ranges, the mechanically and functionally equivalent static RAM, CDP1822 may be used.

The MWS5101 and MWS5101A types are supplied in 22 lead hermetic dual-in-line, sidebrazed ceramic packages (D suffix), in 22 lead dual-in-line plastic packages (E suffix), and in chip form (H suffix).

Ordering Information

PACKAGE	TEMP. RANGE	MWS5101 250ns	MWS5101 350ns	MWS5101A 250ns	MWS5101A 350ns	PKG. NO.
PDIP Burn-In	0^oC to $+70^oC$	MWS5101EL2	MWS5101ELS	MWS5101AEL2	MWS5101AEL3	E22.4
					MWS5101AEL3X	E22.4
SBDIP Burn-In	0^oC to $+70^oC$	-	MWS5101DL3X	-	MWS5101ADL3	D22.4A
						D22.4A

CAUTION: These devices are sensitive to electrostatic discharge; follow proper IC Handling Procedures.

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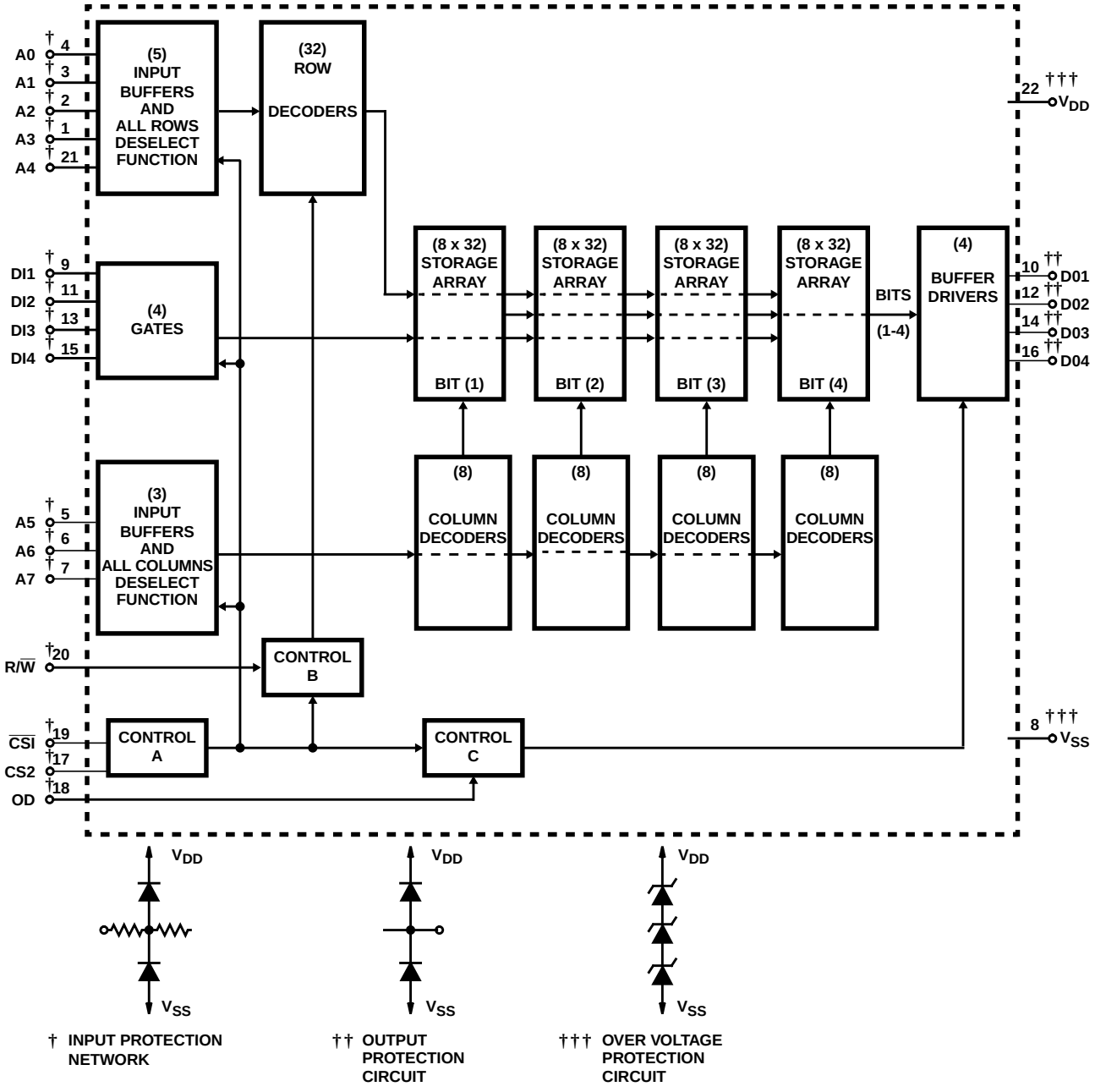
File Number **1106.2**

OPERATIONAL MODES

MODE	INPUTS				OUTPUT
	$\overline{\text{CHIP SELECT 1}}$ ($\overline{CS_1}$)	CHIP SELECT 2 (CS_2)	OUTPUT DISABLE (OD)	READ/$\overline{\text{WRITE}}$ (R/$\overline{W}$)	
Read	0	1	0	1	Read
Write	0	1	0	0	Data In
Write	0	1	1	0	High Impedance
Standby	1	X	X	X	High Impedance
Standby	X	0	X	X	High Impedance
Output Disable	X	X	1	X	High Impedance

NOTE: Logic 1 = High, Logic 0 = Low, X = Don't Care.

Functional Block Diagram



Absolute Maximum Ratings

DC Supply Voltage Range, (V_{DD})
(All Voltages Referenced to V_{SS} Terminal) . . . . . -0.5V to +7V
Input Voltage Range, All Inputs . . . . . -0.5V to V_{DD} +0.5V
DC Input Current, Any One Input. . . . . ±10mA

Thermal Information

	θ_{JA} ($^oC/W$)	θ_{JC} ($^oC/W$)
Thermal Resistance (Typical)		
PDIP Package	75	N/A
SBDIP Package	80	21

Operating Temperature Range (T_A)
Package Type D. . . . . -55^{o}C to +125^{o}C
Package Type E. . . . . -40^{o}C to +85^{o}C
Maximum Storage Temperature Range (T_{STG}) . . . -65^{o}C to +150^{o}C
Maximum Junction Temperature
Ceramic Package . . . . . +175^{o}C
Plastic Package . . . . . +150^{o}C
Maximum Lead Temperature (During Soldering)
At distance 1/16 ±1/32 In. (1.59 ±0.79mm)
from case for 10s max . . . . . +265^{o}C

Recommended Operating Conditions

At T_A = Full Package Temperature Range. For maximum reliability, operating conditions should be selected so that operation is always within the following ranges:

PARAMETER	LIMITS MIN	LIMITS MAX	UNITS
DC Operating Voltage Range	4	6.5	V
Input Voltage Range	V_{SS}	V_{DD}	V

Static Electrical Specifications

At T_A = 0^{o}C to +70^{o}C, V_{DD} = 5V ±5%

PARAMETER		SYMBOL	CONDITIONS V_O (V)	CONDITIONS V_{IN} (V)	LIMITS MWS5101 MIN	LIMITS MWS5101 (NOTE 1) TYP	LIMITS MWS5101 MAX	LIMITS MWS5101A MIN	LIMITS MWS5101A (NOTE 1) TYP	LIMITS MWS5101A MAX	UNITS
Quiescent Device Current	L2 Types	I_{DD}	-	0, 5	-	25	50	-	25	50	µA
	L3 Types		-	0, 10	-	100	200	-	100	200	µA
Output Low (Sink) Current		I_{OL}	0.4	0, 5	2	4	-	2	4	-	mA
Output High (Source) Current		I_{OH}	4.6	0, 5	-1	-2	-	-1	-2	-	mA
Output Voltage Low-Level		V_{OL}	-	0, 5	-	0	0.1	-	0	0.1	V
Output Voltage High-Level		V_{OH}	-	0, 5	4.9	5	-	4.9	5	-	V
Input Low Voltage		V_{IL}	-	-	-	-	1.5	-	-	0.65	V
Input High Voltage		V_{IH}	-	-	3.5	-	-	2.2	-	-	V
Input Leakage Current		I_{IN}	-	0, 5	-	-	±5	-	-	±5	µA
Operating Current (Note 2)		I_{DD1}	-	0, 5	-	4	8	-	4	8	mA
Three-State Output Leakage Current	L2 Types	I_{OUT}	0, 5	0, 5	-	-	±5	-	-	±5	µA
	L3 Types		0, 5	0, 5	-	-	±5	-	-	±5	µA
Input Capacitance		C_{IN}	-	-	-	5	7.5	-	5	7.5	pF
Output Capacitance		C_{OUT}	-	-	-	10	15	-	10	15	pF

NOTES:

1. Typical values are for T_A = +25^{o}C and nominal V_{DD}.
2. Outputs open circuited; Cycle time = 1µs.

Dynamic Electrical Specifications at $T_A = 0^oC$ to $+70^oC$, $V_{DD} = 5V \pm 5\%$

PARAMETER	SYMBOL	LIMITS (NOTE 1)						UNITS
		L2 TYPES			L3 TYPES			
		(NOTE 2) MIN	(NOTE 3) TYP	MAX	(NOTE 2) MIN	(NOTE 3) TYP	MAX	
READ CYCLE TIMES (FIGURE 1)								
Read Cycle	t_{RC}	250	-	-	350	-	-	ns
Access from Address	t_{AA}	-	150	250	-	200	350	ns
Output Valid from $\overline{\text{Chip}}$ $\overline{\text{Select}}$ 1	t_{DOA1}	-	150	250	-	200	350	ns
Output Valid from Chip Select 2	t_{DOA2}	-	150	250	-	200	350	ns
Output Valid from Output Disable	t_{DOA3}	-	-	110	-	-	150	ns
Output Hold from $\overline{\text{Chip}}$ $\overline{\text{Select}}$ 1	t_{DOH1}	20	-	-	20	-	-	ns
Output Hold from Chip Select 2	t_{DOH2}	20	-	-	20	-	-	ns
Output Hold from Output Disable	t_{DOH3}	20	-	-	20	-	-	ns
WRITE CYCLE TIMES (FIGURE 2)								
Write Cycle	t_{WC}	300	-	-	400	-	-	ns
Address Setup	t_{AS}	110	-	-	150	-	-	ns
Write Recovery	t_{WR}	40	-	-	50	-	-	ns
Write Width	t_{WRW}	150	-	-	200	-	-	ns
Input Data Setup Time	t_{DS}	150	-	-	200	-	-	ns
Data in Hold	t_{DH}	40	-	-	50	-	-	ns
$\overline{\text{Chip}}$ $\overline{\text{Select}}$ 1 Setup	$t_{\overline{CS}1S}$	110	-	-	150	-	-	ns
Chip Select 2 Setup	t_{CS2S}	110	-	-	150	-	-	ns
$\overline{\text{Chip}}$ $\overline{\text{Select}}$ 1 Hold	$t_{\overline{CS}1H}$	0	-	-	0	-	-	ns
Chip Select 2 Hold	t_{CS2H}	0	-	-	0	-	-	ns
Output Disable Setup	t_{ODS}	110	-	-	150	-	-	ns

NOTES:

1. MWS5101: t_R, t_F = 20ns, $V_{IH} = 0.7V_{DD}$, $V_{IL} = 0.3V_{DD}$; C_L = 100pF and MWS5101A: t_R, t_F = 20ns, V_{IH} = 2.2V, V_{IL} = 0.65V; C_L = 50pF and 1 TTL Load.
2. Time required by a limit device to allow for the indicated function.
3. Typical values are for $T_A = 25^oC$ and nominal V_{DD}.

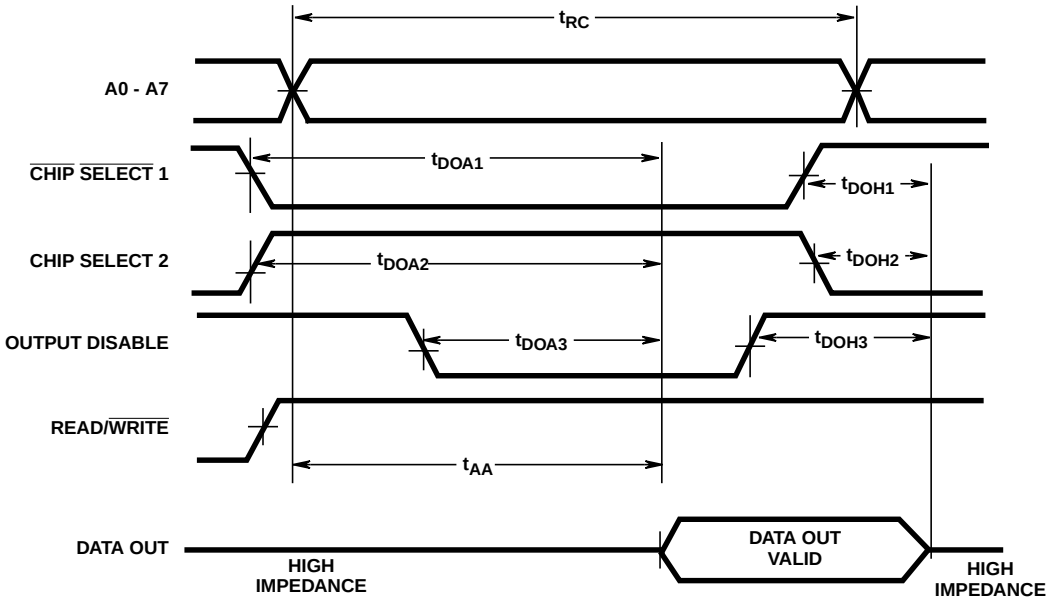
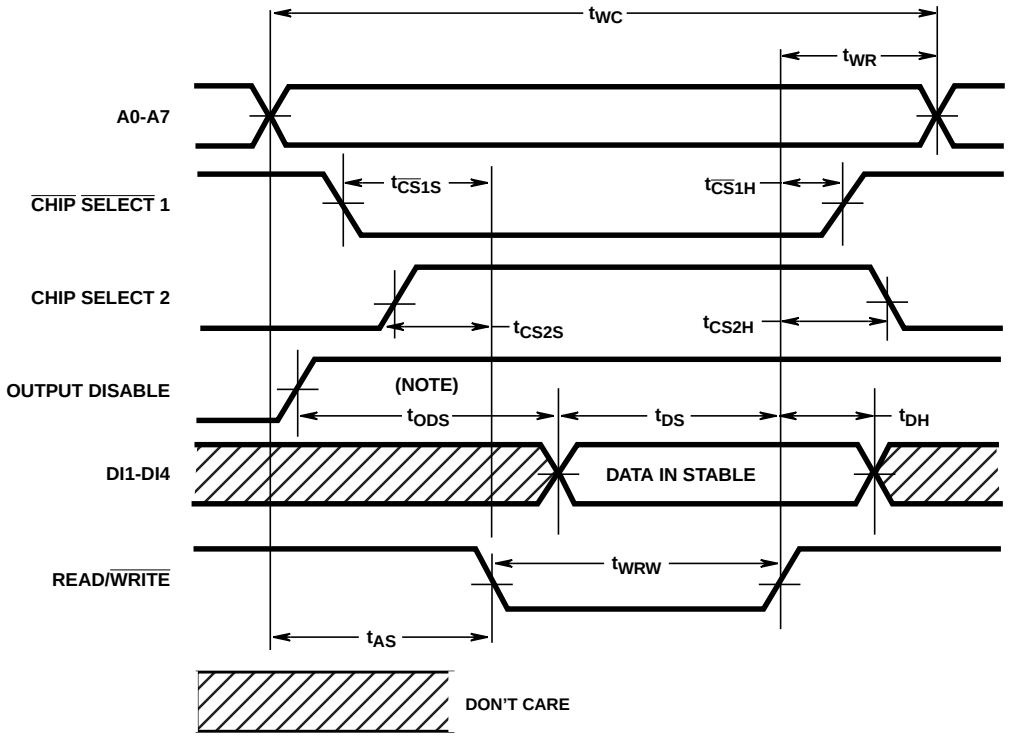


FIGURE 1. READ CYCLE TIMING WAVEFORMS



NOTE: t_{ODS} is required for common I/O operation only; for separate I/O operations, output disable is “don’t care”.

FIGURE 2. WRITE CYCLE TIME WAVEFORMS

Data Retention Specifications at $T_A = 0^oC$ to $+70^oC$; See Figure 3

PARAMETER		SYMBOL	TEST CONDITIONS		LIMITS			
					ALL TYPES			
			V_{DR} (V)	V_{DD} (V)	MIN	(NOTE 1) TYP	MAX	UNITS
Minimum Data Retention Voltage		V_{DR}	-	-	-	1.5	2	V
Data Retention Quiescent Current	L2 Types	I_{DD}	2	-	-	2	10	µA
	L3 Types		2	-	-	5	50	µA
Chip Deselect to Data Retention Time		t_{CDR}	-	5	600	-	-	ns
Recovery to Normal Operation Time		t_{RC}	-	5	600	-	-	ns
V_{DD} to V_{DR} Rise and Fall Time		t_R, t_F	2	5	1	-	-	µs

NOTE:

1. Typical Values are for $T_A = 25^oC$ and nominal V_{DD}.

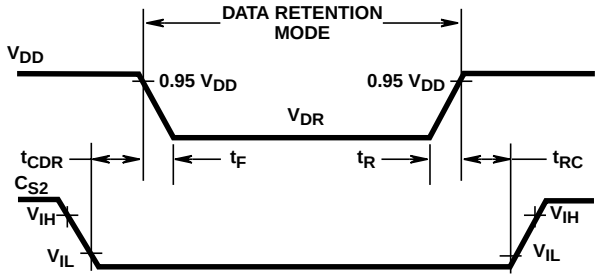


FIGURE 3. LOW V_{DD} DATA RETENTION TIMING WAVEFORMS

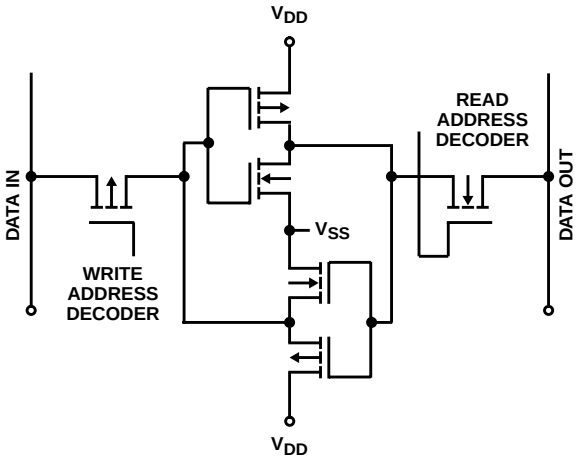


FIGURE 4. MEMORY CELL CONFIGURATION

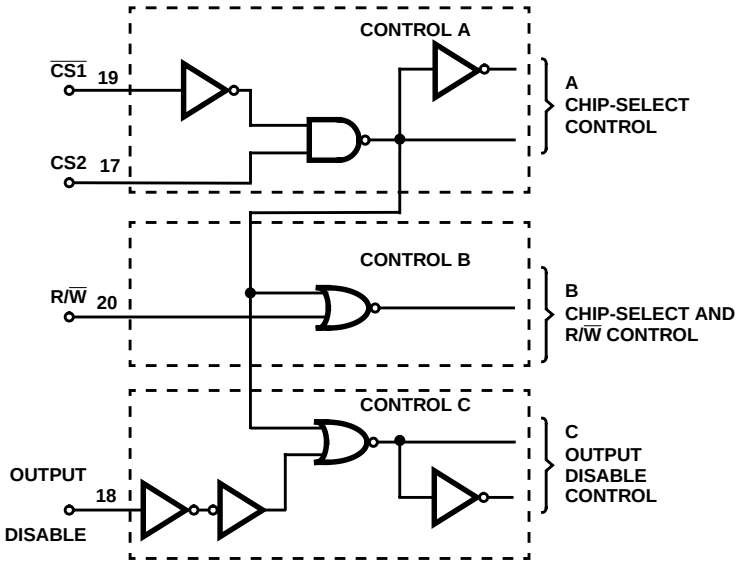


FIGURE 5. LOGIC DIAGRAM OF CONTROLS FOR MWS5101, MWS5101A

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